

Customizable Subscription Box Scheduler

Problem Statement #35 | Retail, E-Commerce & Finance

A subscription box portal where subscribers customize monthly product selections (e.g., books, coffee, snacks) based on preference tags, with support for pausing or skipping billing cycles.

Target Stakeholders / Actors: Subscriber, Fulfillment Lead

Functional Requirements

ID	Description	Priority	Acceptance Criteria	Rationale
FR-001	The system shall allow subscribers to swap monthly box items or pause deliveries up to 48 hours prior to the monthly billing renewal date.	High	Pass: Selection updated and fulfillment manifest refreshed within 5 seconds. Fail: Customization locked out prematurely (before 48-hour window).	Subscribers need flexibility to adjust their box contents based on changing preferences, reducing cancellations and improving satisfaction. The 48-hour cutoff gives the fulfillment team enough lead time to process orders.
FR-002	The system shall allow subscribers to set and update preference tags (e.g., "vegan", "dark roast", "mystery fiction") that drive personalized product recommendations for their monthly box.	High	Pass: Preference tags saved successfully and the recommended product list updates to reflect the new tags on the next page load. Fail: Tags not persisted or recommendations unchanged despite tag updates.	Tag-based personalization is the core value proposition of the platform. Accurate preference matching increases subscriber retention and reduces product return rates.
FR-003	The system shall allow subscribers to skip an upcoming billing cycle without cancelling their subscription, and	Medium	Pass: Billing skipped for the selected cycle, no charge applied, subscription status shows "Skipped", and auto-resumes next cycle.	Allowing skips prevents full cancellations when subscribers temporarily do not need a box (e.g., travel, budget

ID	Description	Priority	Acceptance Criteria	Rationale
	automatically resume the subscription in the following cycle.		Fail: Subscriber charged despite skipping, or subscription does not resume automatically.	constraints), directly improving long-term retention rates.
FR-004	The system shall enable the Fulfillment Lead to view, filter, and export the monthly fulfillment manifest containing subscriber names, shipping addresses, and customized box contents.	High	Pass: Fulfillment manifest displays all active orders for the billing cycle with correct item lists, supports filtering by region/product category, and can be exported as CSV/PDF. Fail: Manifest has missing or incorrect entries, or export fails/produces a corrupted file.	The Fulfillment Lead requires an accurate, exportable manifest to coordinate with warehouse and shipping partners. Errors here lead to wrong shipments and increased operational costs.
FR-005	The system shall send email and in-app notifications to subscribers at least 72 hours before the billing renewal date, reminding them of the upcoming charge and their current box contents with an option to customize.	Medium	Pass: Notification delivered via both email and in-app channels exactly 72 hours before renewal, containing correct box summary and a working link to the customization page. Fail: Notification not sent, sent late (< 48 hours before renewal), or contains incorrect box details.	Timely reminders reduce involuntary churn from unexpected charges and encourage active customization. The 72-hour window precedes the 48-hour customization cutoff from FR-001, giving subscribers ample time.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The fulfillment manifest generator must export shipping labels for 10,000 monthly boxes in under 1	High	Pass: Benchmarking tests confirm manifest generation for 10,000 entries completes in ≤ 60 seconds	The platform must handle a growing subscriber base without degrading the fulfillment

ID	Type	Description	Priority	Acceptance Criteria	Rationale
		minute. All exported data must be transmitted over TLS 1.2+ and access restricted to authenticated Fulfillment Lead accounts only.		under simulated peak load, all API calls use HTTPS/TLS 1.2+, and unauthorized access attempts return HTTP 403. Fail: Generation exceeds 60 seconds, data transmitted over unencrypted channels, or unauthorized users can access the manifest.	workflow. Shipping data contains PII (names, addresses), so performance must not come at the expense of security.
NFR-002	Availability & Scalability	The subscription management portal shall maintain 99.9% uptime on a monthly basis and support horizontal scaling to handle up to 50,000 concurrent subscriber sessions during peak customization windows (48–72 hours before billing renewal).	High	Pass: Monitoring reports confirm ≥ 99.9% uptime over 3 consecutive months, and load tests demonstrate stable response times (< 2 seconds per page) with 50,000 simulated concurrent users. Fail: Monthly uptime falls below 99.9%, or response times exceed 5 seconds under peak simulated load.	The 48-hour customization window creates predictable traffic spikes. System downtime during this critical period directly causes lost customization opportunities and subscriber frustration.